



Framework for Agent Gamification Based on Ontologies

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25 June 2025

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Introduction

Assistant Professor at the University of Zagreb **Faculty of Organization and Informatics**, and a member of the Artificial Intelligence Laboratory at UNIZG FOI.

Scientific interests mainly in:

- multiagent systems,
- semantic modelling,
- gamification,
- artificial intelligence,
- computer games.

One of the teachers of the following courses in Croatian or English:

- Multiagent Systems,
- Database Theory,
- Declarative Programming,
- Introduction to Artificial Intelligence,
- Introduction to Computer Games,
- Internet Security,
- Computer Game Development Platforms.

Engaged in international activities and promoting international relations:

- Erasmus student at **Karl-Franzens University of Graz** (AT),
- Erasmus intern at **Jožef Stefan Institute** in Ljubljana (SI),
- Erasmus+ intern at **Elettra Sincrotrone** in Trieste (IT),
- 3-month research stay at **Universitat Politècnica de València** in Valencia (ES),
- ITEC student at **Centre for Development of Advanced Computing** in NOIDA (IN),
- 16-month research visit at **Universitat Politècnica de València** in Valencia (ES),

Introduction

Publications



Project activity



BSc and MA Mentees



Overview

	Part I	Part II
Phase A	MAS ontology	gamification ontology
Phase B	MAS framework	gamification framework

Table 1: Research in a nutshell: two parts, two phases each

- Phase 2 focuses on:
 - GIVEN - Games as Intelligent Virtual Environments Ontology
 - MAGO-Ga - Gamification of an Intelligent Virtual Environment Ontology
 - MAGO-Ga Framework - Implementation and case study

- METHONTOLOGY ontology engineering methodology
- Structured approach with clearly defined steps:
 - Specification
 - Knowledge Acquisition
 - Conceptualisation
 - Formalisation
 - Integration
 - Implementation and Evaluation
- Applied to both GIVEn and MAGO-Ga ontologies

- **Purpose:** Provide concepts for describing video games as cro:IVEintelligent virtual environments (IVEs)
- **Intended use:** Modelling video game worlds using applicable concepts
- **Applications:**
 - Simulations
 - Enhancing reasoning capabilities of intelligent agents
 - Content generation (worlds, NPCs, items)
- **Level of granularity:** Abstract concepts that can be further specialized

- Analysis of existing video game ontologies:
 - Core Game Ontology (CGO)
 - Video Game Ontology (VGO)
- Concepts identified for modelling video games as IVEs:
 - NPCs, monsters, mobs
 - Places, points of interest
 - Game mechanics and dynamics

- Integration with existing ontologies:
 - MAMbO5 - multiagent systems ontology
 - CGO - video game ontology
 - VGO - video game ontology
- Benefits:
 - Enhanced semantic reach
 - Improved reusability
 - Data consistency
 - Reduced redundancy

- **Purpose:** Model gamified multi-agent systems as IVEs
- **Key components:**
 - Formal representation of gamification elements
 - Personality traits based on Five Factor model
 - Interactions between agents and gamification elements
- **Intended use:** Knowledge base for reasoning about gamified multi-agent systems

- GEAR - Gamified Emergent Agent Relations
- Implementation based on the recipeWorld model
- Key elements:
 - Service providers and consumers
 - Gamification techniques to influence agent behaviour
 - Personality profiles based on Five-Factor Model
 - Compatibility calculation between gamification techniques and agents
- Available on GitHub: <https://github.com/AILab-FOI/GEAR>

Case Study: The Recipe World

- Based on the recipeWorld model
- Agents as service providers and consumers
- Gamification techniques influence agent behaviour
- Personality profiles based on Five-Factor Model
- Compatibility between gamification techniques and agents

Implementation

- Both ontologies implemented using Protégé
- Available on GitHub repository
- OWL/XML serialization in appendices
- Consistent and provides additional knowledge when reasoned upon
- GEAR framework implements the concepts in a practical application

- Ontologies evaluated by domain experts
- Interview and review approaches
- Tested against specification documents
- Framework demonstrated through case study

Conclusion

- Phase 2 delivered:
 - GIVEn - Ontology for modelling video games as IVEs
 - MAGO-Ga - Ontology for gamified multi-agent systems
 - GEAR - Framework implementing the concepts
- Contributions:
 - Formal representation of gamification in multi-agent systems
 - Integration of personality traits with gamification techniques
 - Practical implementation demonstrating the approach

Bibliography

Acknowledgement

Acknowledgement



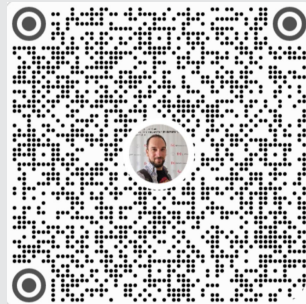
**Funded by
the European Union**
NextGenerationEU

MOBODL-2023-08-5618

This work was funded by the European Union and the Croatian Science Foundation.



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